

physics, and the one before that was perhaps in the medical school. This kind of erudition impressed my father far more than scientific assembly-line work. I may have something in my genes driving me away from *bildungsphilisters*.

Although Mandelbrot often expressed amazement at the temperament of high-flying erudites and remarkable but not-so-famous scientists, such as his old friend Carleton Gajdusek, a man who impressed him with his ability to uncover the causes of tropical diseases, he did not seem eager to trumpet his association with those we consider great scientists. It took me a while to discover that he had worked with an impressive list of scientists in seemingly every field, something a name-dropper would have brought up continuously. Although I have been working with him for a few years now, only the other day, as I was chatting with his wife, did I discover that he spent two years as the mathematical collaborator of the psychologist Jean Piaget. Another shock came when I discovered that he had also worked with the great historian Fernand Braudel, but Mandelbrot did not seem to be interested in Braudel. He did not care to discuss John von Neuman with whom he had worked as a postdoctoral fellow. His scale was inverted. I asked him once about Charles Tresser, an unknown physicist I met at a party who wrote papers on chaos theory and supplemented his researcher's income by making pastry for a shop he ran near New York City. He was emphatic: "*un homme extraordinaire*," he called Tresser, and could not stop praising him. But when I asked him about a particular famous hotshot, he replied, "He is the prototypical *bon élève*, a student with good grades, no depth, and no vision." That hotshot was a Nobel laureate.

THE PLATONICITY OF TRIANGLES

Now, why am I calling this business Mandelbrotian, or fractal, randomness? Every single bit and piece of the puzzle has been previously mentioned by someone else, such as Pareto, Yule, and Zipf, but it was Mandelbrot who a) connected the dots, b) linked randomness to geometry (and a special brand at that), and c) took the subject to its natural conclusion. Indeed many mathematicians are famous today partly because he dug out their works to back up his claims—the strategy I am following here in this book. "I had to invent my predecessors, so people take me seriously," he once told me, and he used the credibility of big guns as a rhetorical device. One can almost always ferret out predecessors for any thought. You can always find someone who worked on a part of your argument and use his contribution as your backup. The scientific association with a big idea, the "brand name," goes to the one who connects the dots, not the one who makes a casual observation—even Charles Darwin, who uncultured scientists claim "invented" the survival of the fittest, was not the first to mention it. He wrote in the introduction of *The*